

# LECTURE NOTES MATH254

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## 1. LECTURE 3

In this class we introduce subrings, subfields, homomorphisms and isomorphisms.

**1.1. Subrings and subfields.** As in the case of groups we have the notion of *subrings* and *subfields*.

**Definition 1.1.** Let  $R$  be a ring. A subring  $S$  of  $R$  is a subset  $S \subset R$  such that  $S$  is closed under both  $+$  and  $\cdot$ , and such that  $S$  is a ring with the operations  $+$  and  $\cdot$ .

**Definition 1.2.** Let  $F$  be a field. A subfield  $L$  of  $F$  is a subset  $L \subset F$  such that  $L$  is closed under both  $+$  and  $\cdot$ , and such that  $L$  is a field with the operations  $+$  and  $\cdot$ .

**Remark 1.** We also have the notions like *proper subrings* and *proper subfields* which are defined completely analogously as in the case of groups.

**Example 1.1.** The ring  $\mathbb{Z}$  is a subring of  $\mathbb{Q}$ . The field  $\mathbb{Q}$  is a subfield of  $\mathbb{R}$  (and  $\mathbb{C}$ ).

**Example 1.2.** The ring  $C_0^\infty(\mathbb{R})$  is a subring of the ring  $C^\infty(\mathbb{R})$ . Note that  $C^\infty(\mathbb{R})$  is a ring with unity, while  $C_0^\infty(\mathbb{R})$  is not a ring with unity, so a subring of a ring with unity is not necessarily a ring with unity.

**1.2. Homomorphisms and isomorphisms of rings and fields.** When we studied groups in the previous course a very important concept was that of a *homomorphism* and an *isomorphism*. Similar objects can be defined in the context of rings and fields.

**Definition 1.3.** Let  $R$  and  $S$  be rings. A map  $\phi : R \rightarrow S$  is a homomorphism (of rings) if for all  $a, b \in R$ :

- (i)  $\phi(a + b) = \phi(a) + \phi(b)$ ,
- (ii)  $\phi(ab) = \phi(a)\phi(b)$ .

**Definition 1.4.** Let  $R$  and  $S$  be rings. A map  $\phi : R \rightarrow S$  is an isomorphism if  $\phi$  is a homomorphism and  $\phi$  is bijective.

A homomorphism between fields is a homomorphism between fields is a homomorphism between the underlying rings. However, here we can add an extra assumption, that the multiplicative identity (the unity) should be mapped to the multiplicative identity. This excludes the trivial map as a homomorphism, which is often useful (but not everyone use this convention).

**Definition 1.5.** Let  $F$  and  $L$  be fields. A map  $\phi : F \rightarrow L$  is a homomorphism (of fields) if for all  $a, b \in F$  we have:

- (1)  $\phi(a + b) = \phi(a) + \phi(b)$ ,
- (2)  $\phi(ab) = \phi(a)\phi(b)$ ,
- (3) if  $1_F$  is unity in  $F$  and  $1_L$  is unity in  $L$  then  $\phi(1_F) = 1_L$ .

**Definition 1.6.** Let  $F$  and  $L$  be fields. A map  $\phi : F \rightarrow L$  is an isomorphism if  $\phi$  is a homomorphism and  $\phi$  is bijective.

**Remark 2.** If  $R$  and  $S$  are rings (or fields) and there is an isomorphism  $\phi : R \rightarrow S$  then we say that  $R$  and  $S$  are isomorphic.

**Lemma 1.1.** If  $\phi : R \rightarrow S$  is an isomorphism of rings (or fields) then  $\phi^{-1} : S \rightarrow R$  is also an isomorphism of rings (or fields).

*Proof.* Since  $\phi$  is bijective,  $\phi^{-1}$  is also bijective, so it remains to show that  $\phi^{-1}(x+y) = \phi^{-1}(x) + \phi^{-1}(y)$  and  $\phi^{-1}(xy) = \phi^{-1}(x)\phi^{-1}(y)$  for all  $x, y \in S$  (and in the case of fields we also need to show  $\phi^{-1}(1_S) = 1_R$ ). Let  $x, y \in S$ , using that  $\phi$  is a homomorphism we obtain:

$$(1.1) \quad \phi(\phi^{-1}(x) + \phi^{-1}(y)) = \phi(\phi^{-1}(x)) + \phi(\phi^{-1}(y)) = x + y$$

so after applying  $\phi^{-1}$  on both sides we obtain  $\phi^{-1}(x + y) = \phi^{-1}(x) + \phi^{-1}(y)$ . The proof that  $\phi^{-1}(xy) = \phi^{-1}(x)\phi^{-1}(y)$  is completely analogous (prove it!!).

It remains to show that, when  $R$  and  $S$  are fields, we have  $\phi^{-1}(1_S) = 1_R$ . However, by definition of a homomorphism of fields we have  $\phi(1_R) = 1_S$ , so after applying  $\phi^{-1}$  on both sides we obtain  $\phi^{-1}(1_S) = 1_R$ . ■

**Example 1.3.** We noted that  $\mathbb{Z}_p$  are examples of fields when  $p$  is prime. If  $n \geq 2$  is an integer (not necessarily prime), then  $\mathbb{Z}_n$  is a ring (with addition and multiplication defined as for  $\mathbb{Z}_p$ ). Note, however, that  $\mathbb{Z}_n$  is not a field unless  $n$  is prime. The map  $\phi : \mathbb{Z} \rightarrow \mathbb{Z}_n$  defined by  $\phi(x) = x + n\mathbb{Z}$  is a homomorphism (prove this, it is essentially immediate from the way we defined addition and multiplication in  $\mathbb{Z}_n$ ).

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<sup>1</sup>Actually, when  $\phi$  is an isomorphism of fields then  $\phi$  can not map every element to 0 in  $S$  since a field contains at least two elements (0 and 1 since  $0 \neq 1$  is part of the definition), which implies that  $\phi(1_R) = \phi(1_R 1_R) = \phi(1_R)\phi(1_R)$  and after dividing by  $\phi(1_R)$  (note that  $\phi(0_R) = 0_S$ , so  $\phi(1_R) \neq 0_S$ ) we obtain  $\phi(1_R) = 1_S$ . That is, for isomorphisms the assumption that  $\phi(1_R) = 1_S$  in the definition of homomorphisms of fields is redundant. The only reason we add  $\phi(1_R) = 1_S$  is to guarantee that  $\phi$  does not map every element to  $0_S$ .

**Example 1.4.** Recall that as abelian groups  $2\mathbb{Z}$  and  $3\mathbb{Z}$  are isomorphic. This might make one believe that  $2\mathbb{Z}$  and  $3\mathbb{Z}$  should be isomorphic as rings as well, however this is **not** true. In fact, any homomorphism  $\phi : 2\mathbb{Z} \rightarrow 3\mathbb{Z}$  is constant equal to 0 (so it can not possibly be invertible). To see this, let  $\phi : 2\mathbb{Z} \rightarrow 3\mathbb{Z}$  be any homomorphism. We use the axioms for homomorphisms to calculate  $\phi(4)$  in two different ways. First, let  $\phi(2) = 3n$  for some  $n \in \mathbb{Z}$ . We then have:

$$(1.2) \quad \phi(4) = \phi(2 \cdot 2) = \phi(2)\phi(2) = 9n^2,$$

$$(1.3) \quad \phi(4) = \phi(2 + 2) = \phi(2) + \phi(2) = 3n + 3n = 6n.$$

So we must have  $9n^2 = 6n$  or after rearranging a bit:  $3n(3n - 2) = 0$ . Note that  $3n - 2 \neq 0$  for all integers  $n$  (since 2 is prime), so if  $3n(3n - 2) = 0$  then  $3n = 0$ . But if  $3n = 0$  then  $n = 0$ . It follows that  $\phi(2) = 0$ . We leave the remainder of the proof that  $\phi$  is constant equal to 0 as an exercise (prove this! Note that  $\phi(-2) = 0$  since  $0 = \phi(0) = \phi(2 - 2) = \phi(-2) + 0$ , and deal with any  $2m \in 2\mathbb{Z}$  by induction on  $m$ ).

As in the case of homomorphisms of groups we can also define kernels of homomorphisms:

**Definition 1.7.** Let  $\phi : R \rightarrow S$  be a homomorphism of rings. We define the kernel of  $\phi$  by:

$$(1.4) \quad \ker \phi = \{a \in R \mid \phi(a) = 0_S\}.$$

**Remark 3.** We could also define kernels of homomorphisms of fields, however this would not be very interesting. If  $\phi : F \rightarrow L$  is a homomorphism of fields and if  $a \in F$  is such that  $\phi(a) = 0_L$  then  $a = 0_F$ . Indeed, if  $a \neq 0_F$  then we have some inverse  $a^{-1} \in F$  and we would have  $\phi(1_F) = \phi(aa^{-1}) = \phi(a)\phi(a^{-1}) = 0_L \cdot \phi(a^{-1}) = 0_L$  (where we used that  $0_L \cdot a = 0_L$  for all  $a \in L$ ). But this can not be the case since, by definition, we have  $\phi(1_F) = 1_L$  and  $0_L \neq 1_L$ . So we conclude that  $\phi(a) = 0_L$  implies  $a = 0_F$ . This actually implies that any homomorphism of fields is injective (prove this!).

**Lemma 1.2.** If  $\phi : R \rightarrow S$  is a homomorphism of rings, then  $\ker \phi$  is a subring of  $R$ .

*Proof.* First we show that  $\ker \phi$  is closed under  $+$  and  $\cdot$ . Let  $a, b \in \ker \phi$ , then:

$$\phi(a + b) = \phi(a) + \phi(b) = 0 + 0 = 0,$$

$$\phi(ab) = \phi(a)\phi(b) = 0 \cdot 0 = 0,$$

which proves that  $a + b, ab \in \ker \phi$ . That is,  $\ker \phi$  is closed under  $+$  and  $\cdot$ . Next we prove that  $(\ker \phi, +)$  is an abelian group. We already know that  $\ker \phi$  is closed under  $+$  so we must only prove that  $0_R \in \ker \phi$  and if  $a \in \ker \phi$  then  $-a \in \ker \phi$ . First, note that  $\phi(0_R) = \phi(0_R + 0_R) = \phi(0_R) + \phi(0_R)$  so after subtracting  $\phi(0_R)$  on both sides we obtain  $0_S = \phi(0_R)$ , that is  $0_R \in \ker \phi$ . Second, if  $a \in \ker \phi$  then  $0_S = \phi(0_R) = \phi(a - a) = \phi(a) + \phi(-a) = 0_S + \phi(-a)$  or  $\phi(-a) = 0_S$ . That is, if  $a \in \ker \phi$  then  $-a \in \ker \phi$ , and this finishes the proof that  $(\ker \phi, +)$  is an abelian group. It remains to show that  $\cdot$  is associative in  $\ker \phi$  and that it satisfies the distributive law, however this is immediate since the same thing holds in  $R$ . ■

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